

Blood pressure measurement

Shna Wirya Hawez
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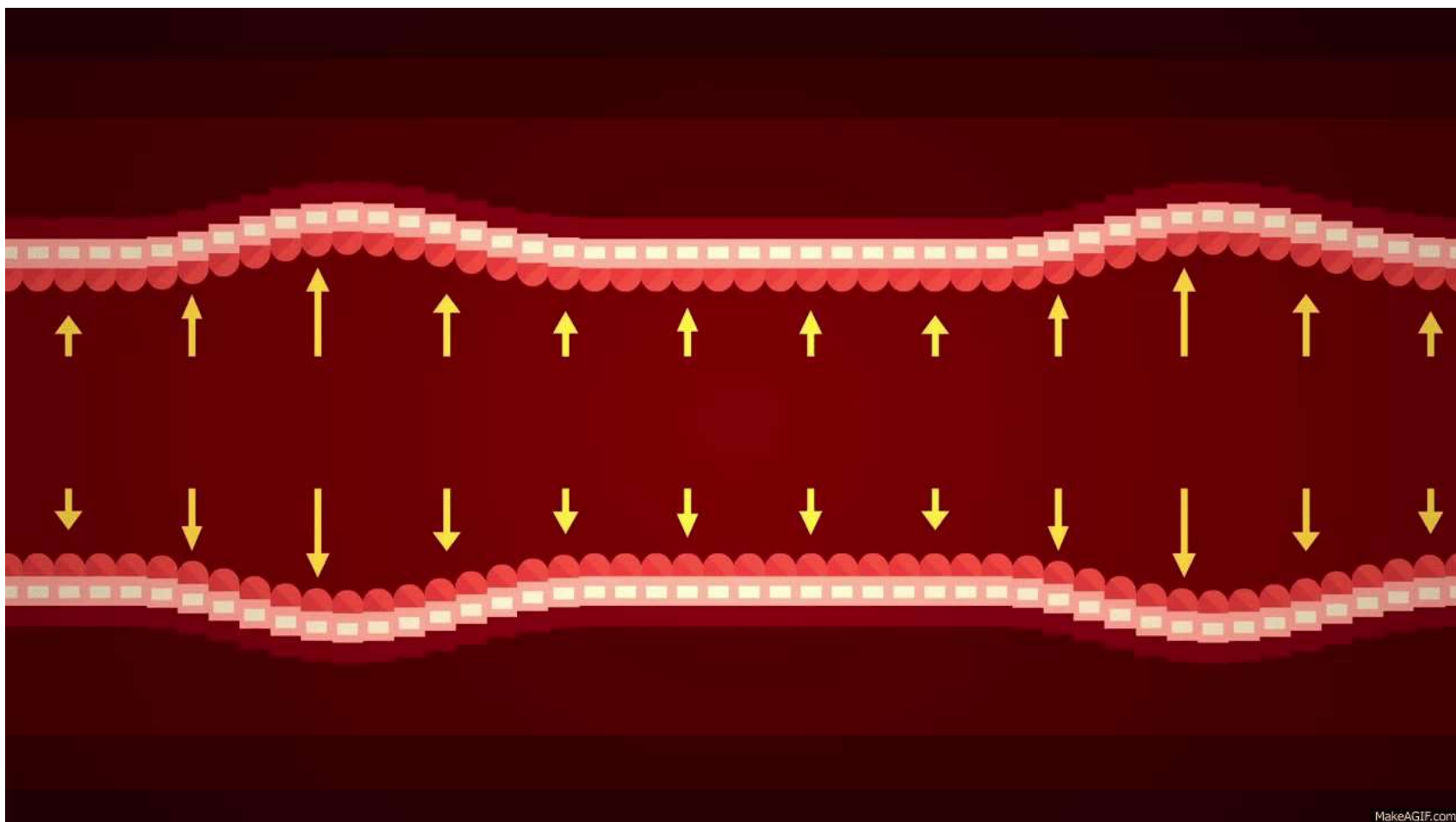


The learning out comes

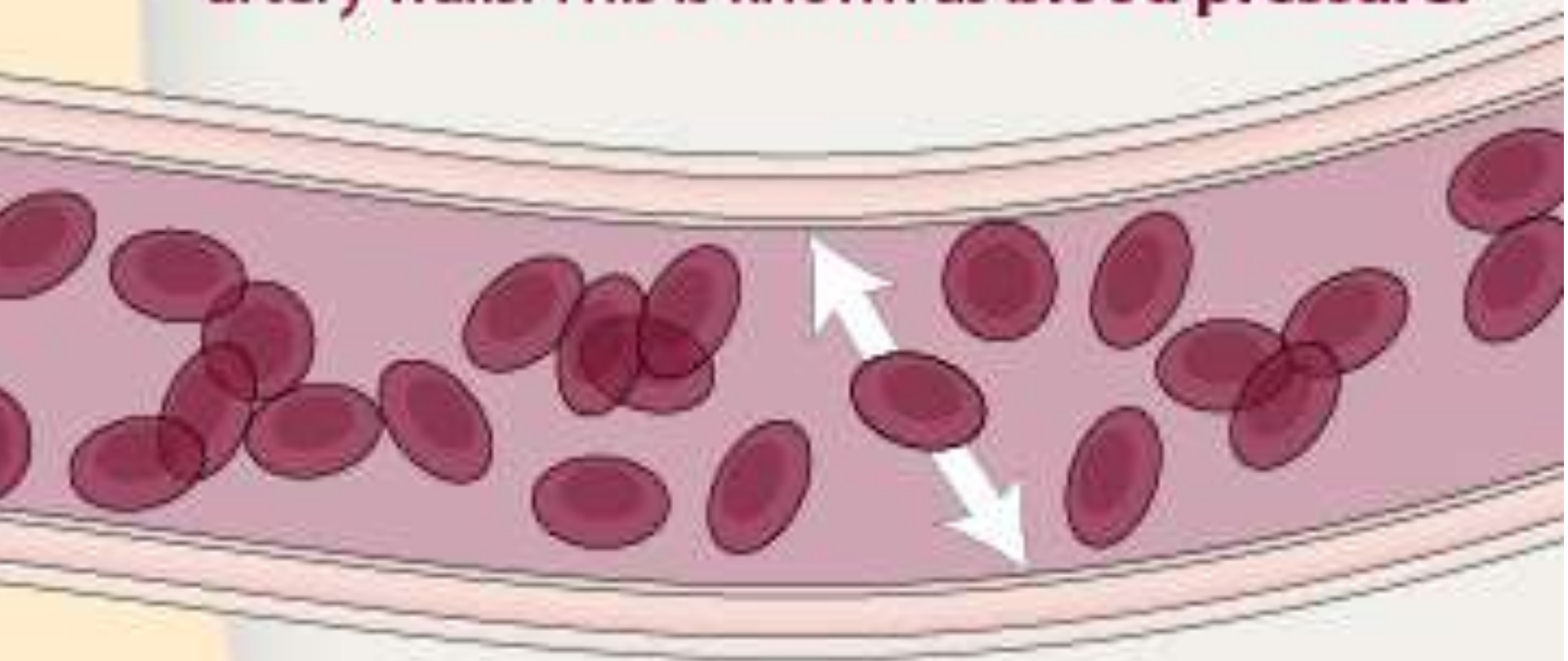
- **By the end of this session the student will be able :**
- What is blood pressure?
- Physical principle of blood pressure.
- What do blood pressure numbers mean?
- Type of blood pressure measuring tools
- procedure

What is blood pressure

- Blood pressure is the pressure of blood pushing against the walls of your arteries. Arteries carry blood from your heart to other parts of your body.
- Your blood pressure normally rises and falls throughout the day.



When the heart pumps blood through the arteries, the blood puts pressure on the artery walls. This is known as **blood pressure**.



Physical principle of blood pressure

The diagram shows the formula $P = \rho gh$ on a light blue background. Three labels with leader lines point to the variables: 'the density of the liquid' points to ρ , 'the acceleration of gravity' points to g , and 'the height of the liquid (or depth of the fluid)' points to h .

$$P = \rho gh$$

the density of the liquid

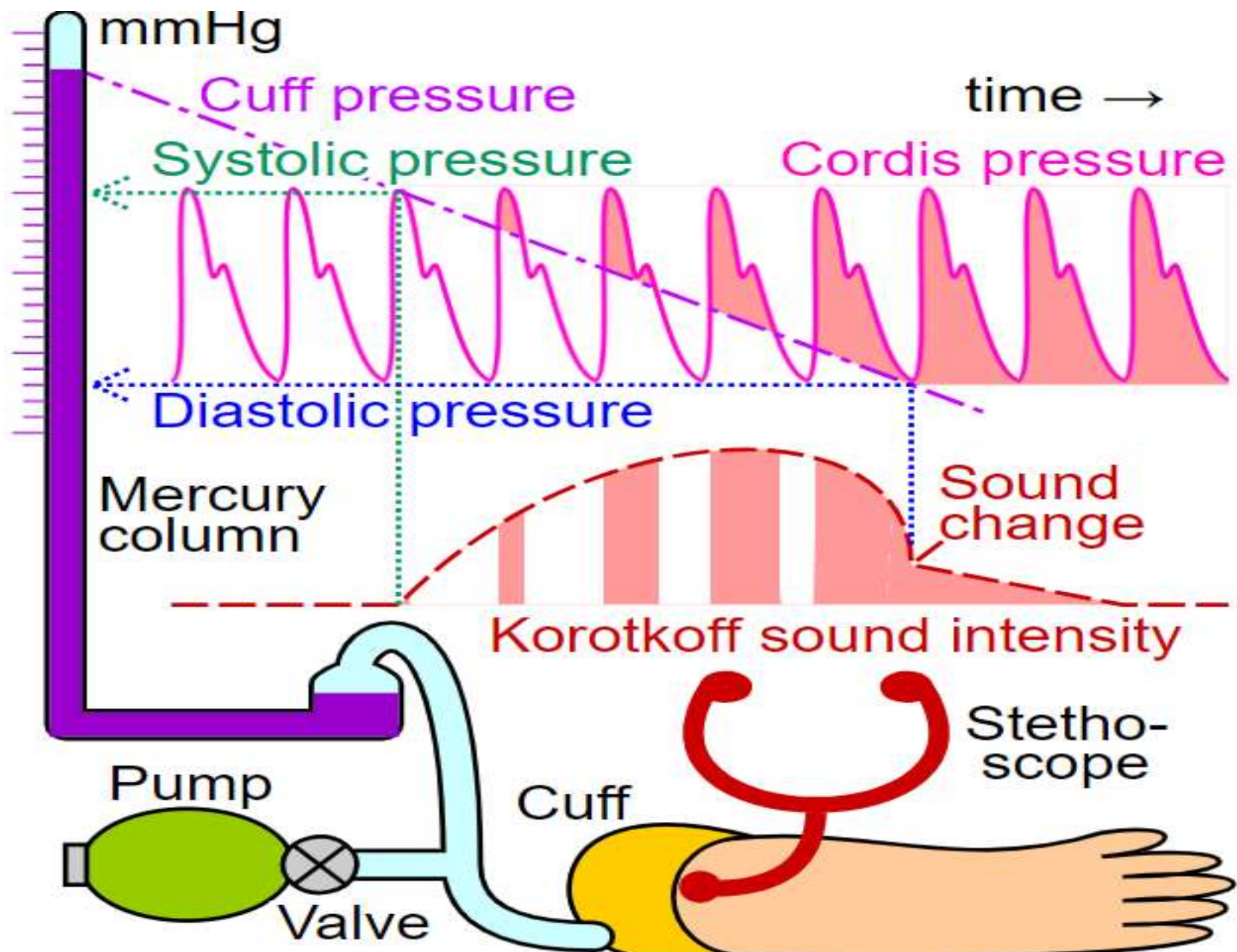
the acceleration of gravity

the height of the liquid (or depth of the fluid)

What do blood pressure numbers mean?

- Blood pressure is measured using two numbers:
- The first number, called *systolic* blood pressure,.
- The second number, called *diastolic* blood pressure,

- **What Are Systolic and Diastolic Blood Pressure?**
- **What Is a Normal Blood Pressure?**



Korotkoff sounds

- The Korotkoff sounds are the loud thumping sounds you'll hear through the stethoscope while deflating the sphygmomanometer.

What Are Systolic and Diastolic Blood Pressures?



120

—

80

Systolic Blood Pressure

- Pressure exerted when blood is ejected into arteries
- Normal systolic blood pressure is 120 mmHg or below

Diastolic Blood Pressure

- Pressure blood exerts within arteries between heartbeats
- Normal diastolic blood pressure is 80 mmHg or below

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***systolic* blood pressure,**

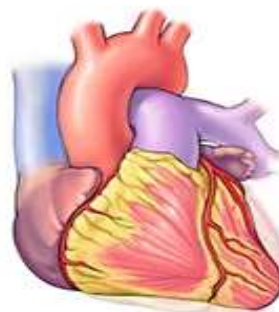
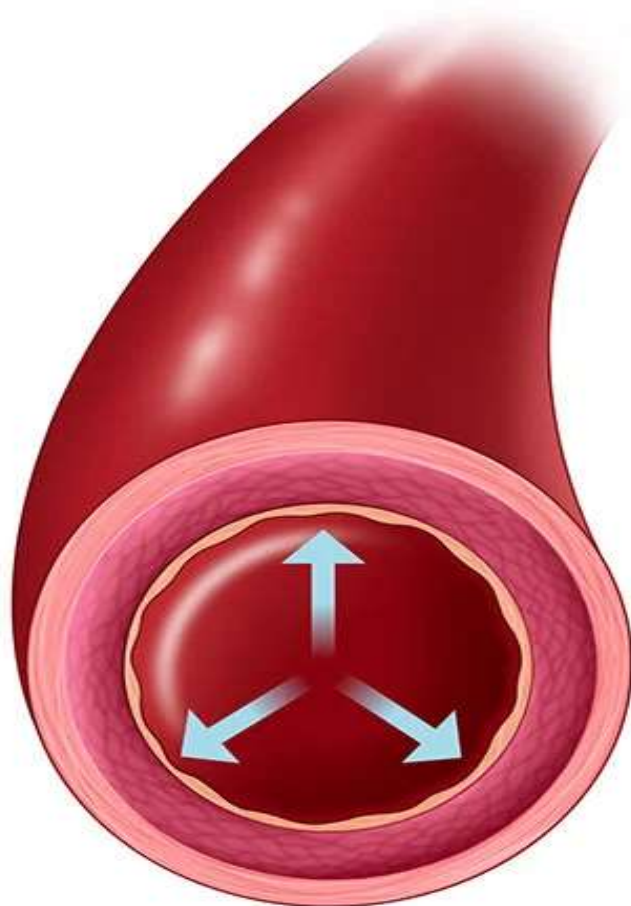
measures the pressure in your arteries when
your heart beats

***diastolic* blood pressure,**

measures the pressure in your arteries when your
heart rests between beats

High Blood Pressure

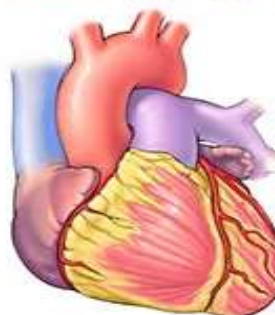
Hypertension



≥ 130

Systolic number
is when the
heart contracts

≥ 80



Diastolic number
is when the
heart rests

■ Force of blood on artery wall

Type of blood pressure measuring tools

SPHYGMOMANOMETER



Has a calibrated glass tube containing mercury

ANEROID



Has a calibrated dial with a needle that points to numbers on the face of the dial

ELECTRONIC / DIGITAL

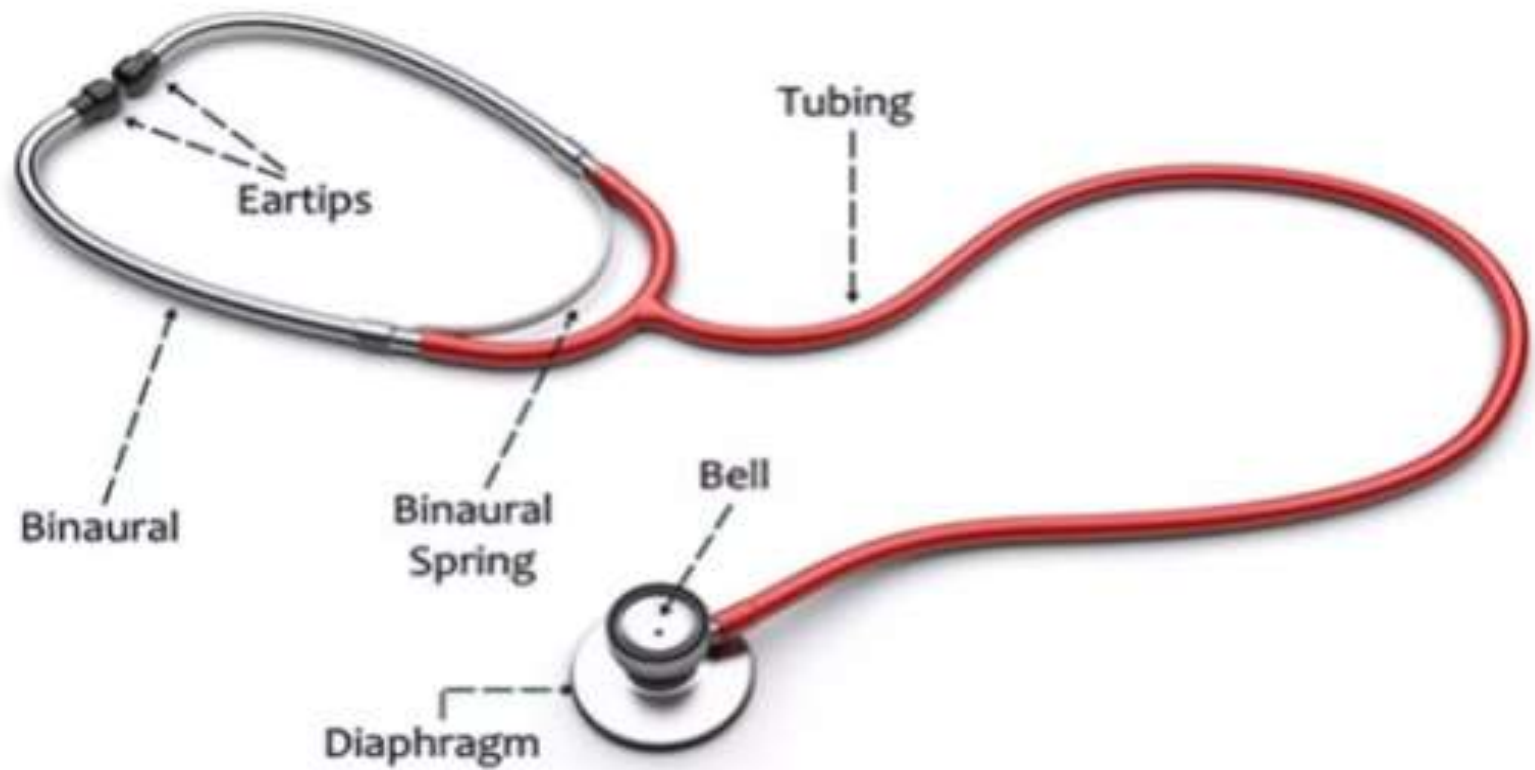


Uses a digital display and usually includes the pulse rate

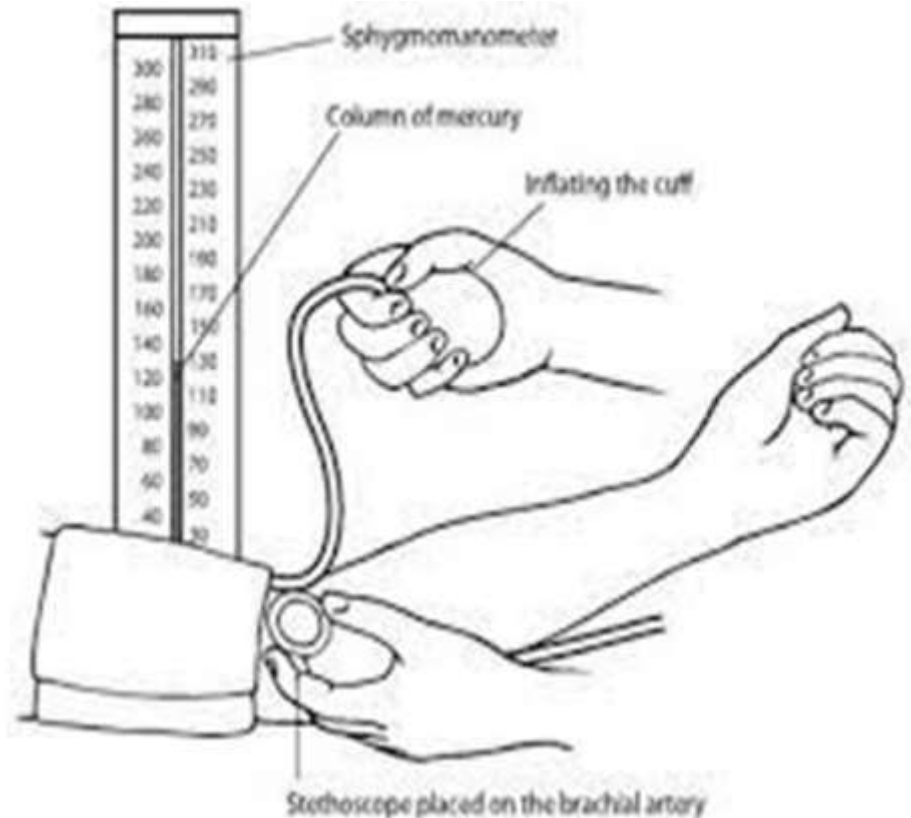
procedure

- Ensure there is no air trapping in the cuff.
- Place the manometer at the heart level.
- Wrap the cuff evenly and firmly around upper arm.
- Ensure manometer is positioned straight and at eye level.
- Deflate the cuff fully.
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- Clean stethoscope eartips and diaphragm, place eartips into ears and ensure sound are clear and not muffled.



- Place diaphragm over the brachial pulse .
- Slowly deflate cuff and listen to both systolic & diastolic reading.



Manual Blood Pressure Monitor: A Step-by-Step Guide to Accurately Measuring Blood Pressure



Summary

- Blood pressure is a measure of the force that your heart uses to pump blood around your body.
- A sphygmomanometer has three parts: a cuff that can be inflated with air, a pressure meter (manometer) for measuring air pressure in the cuff, and. a stethoscope for listening to the sound the blood makes as it flows through the brachial artery (the major artery found in your upper arm)

Assignment

- **What is the correct way to measure blood pressure?**

References

- <https://www.slideshare.net/cikbungazafieyaha/mid/blood-pressure>.
- [National Center for Chronic Disease Prevention and Health Promotion](#) , [Division for Heart Disease and Stroke Prevention](#)

Thank you